

Harry Chandra Tsjan

harrychandratsjan@gmail.com | +852-54640460 | github.com/h4r1yzz |
linkedin.com/in/harry-chandra-tsjan

Education

Hong Kong University of Science and Technology, MSc. Information Technology	Sep 2026 – Sep 2027
University Tunku Abdul Rahman, B.Sc. in Software Engineering	Oct 2021 – Oct 2024

Experience

AI Engineer, Carbon AI	Dec 2025 – Aug 2026
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- Built and enhanced production AI applications using Python, FastAPI, LangChain, and LangGraph, implementing multi-agent workflows and LLM-powered application capabilities.
- Developed a hybrid Retrieval-Augmented Generation (RAG) pipeline using pgvector, Vertex AI embeddings, and semantic retrieval techniques, improving document retrieval accuracy by approximately 25%.
- Architected and managed cloud-native infrastructure on Google Cloud Platform, including Cloud Run, Cloud SQL/PostgreSQL, Pub/Sub background workers, and supporting production services.

Data Science, RosaryLabs	Mar 2025 – Aug 2025
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- Designed an MCP-based search and agent workflow enabling dynamic tool selection and multi-turn reasoning for complex information-retrieval tasks.
- Built evaluation pipelines using DeepEval and custom LLM metrics to assess response quality and identify failure cases across AI workflows.
- Evaluated Qdrant vector search and sparse embedding approaches for Retrieval-Augmented Generation, comparing retrieval strategies for relevance and search quality.

Projects

NeuroGrade | Python, LangChain, LLM, React, FastAPI

- Built a workflow that allows clinicians to upload four supported H5 brain-imaging inputs, then uses machine learning and similarity search to retrieve comparable historical images based on tumor location and imaging characteristics.
- Developed an AI agent pipeline that consolidates retrieved imaging evidence and generated findings into a structured view, enabling clinicians to review the results and provide their own clinical assessment.
- Automated structured report generation by combining AI-assisted findings, retrieved imaging evidence, and the clinician's final assessment, improving workflow efficiency while preserving clinician oversight and reliability.

Technologies

Programming: Python, SQL, NoSQL, C#, Java, JavaScript, Dart, Spark

Technologies: MSSQL, PostgreSQL, AWS EC2, Git, Azure Data Studio, Tableau, Snowflake, Docker, Azure, Airflow, dbt

Languages: English (Fluent), Chinese (Fluent), Malay (Fluent)

Awards & Certifications

AWS Cloud Practitioner Foundational	Dec 2024
NVIDIA Fundamentals of Deep Learning	Dec 2024
First Runner-Up, ASEAN AI Malaysia Summit Buildathon	Oct 2025